

Tree Traversals

Tree Traversal means visiting each node exactly once in a tree

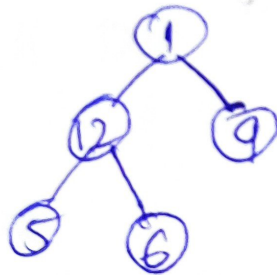
Types:-

-> Inorder tree traversal (LNR)

-> Preorder " " (NLR)

-> Post order " " (LRN)

Constructing a Binary Tree for 1, 12, 9, 5, 6



~~NLR~~ ->

LRN -> H

NLR ->

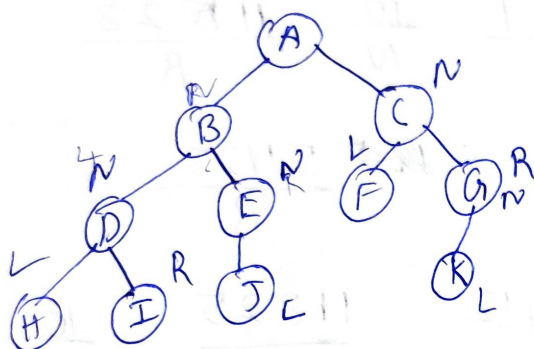
$$1) LNR \rightarrow \frac{5 \ 12 \ 6}{L} \quad \frac{1}{N} \quad \frac{9}{R}$$

$$2) NLR \rightarrow \frac{1}{N} \quad \frac{12 \ 5 \ 6}{L} \quad \frac{9}{R}$$

$$3) LRN \rightarrow \frac{5 \ 6 \ 12}{L} \quad \frac{9}{R} \quad \frac{1}{N}$$

example

Find the inorder



$$LNR \rightarrow \frac{H \ D \ I}{L} \quad \frac{B}{N} \quad \frac{J \ E}{R} \quad \frac{A}{N}$$

$$\frac{F \ C \ K \ G}{R}$$

$$\cancel{NLR} \rightarrow$$

$$LRN \rightarrow \frac{H \ I \ D \ J \ E \ B}{L} \quad \frac{F \ K \ G \ C}{R} \quad \frac{A}{N}$$

$$NLR \rightarrow \frac{A}{N} \quad H \ I \ D \ B \ C$$

NLR $\rightarrow \frac{A}{N}$

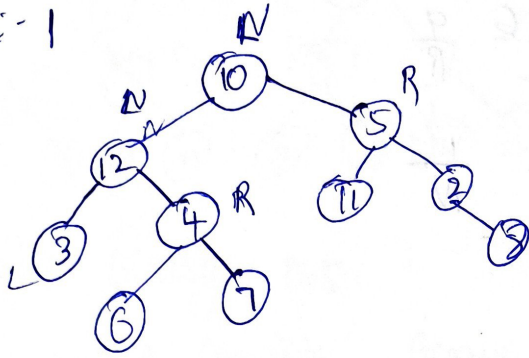
BDHIEJC
L

CFGE
R

Name: AK
Reg No: 1925

31/7/26

Ex:-1

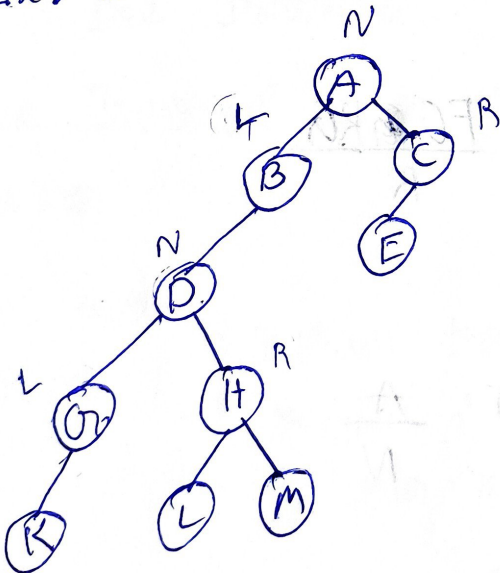


LNR $\rightarrow$ ~~3~~ 3 1 2 6 4 7 $\frac{10}{N}$ 11 2 8
L R

NLR $\rightarrow \frac{10}{N}$ 1 2 3 4 6 7 ~~5~~ 5 1 1 2 8
L R

LRN $\rightarrow$ ~~3 1 2 6 4 7~~ 3 6 7 4 1 2 11 8 2 5 $\frac{10}{N}$
L R

Ex:-2



LNR $\rightarrow$ K G D L H M ~~B C~~ $\frac{A}{N}$ E C
L R

LRN $\rightarrow$ K G L M H D B ~~A C~~ $\frac{E C}{R}$ $\frac{A}{N}$
L

NLR $\rightarrow \frac{A}{N}$

high

Balancing

height

Rotation

$\rightarrow$ Rotation

to left or right

Rotation

Example:

Bf = 2-1

Bf = 1

NLR $\rightarrow$

$$\frac{A}{N}$$

B D G K H L M ~~B~~
L

$$\frac{E E}{R}$$